

WASTEWISE: DEVELOPMENT OF A WEB AND MOBILE-BASED WASTE COLLECTION MONITORING AND MANAGEMENT SYSTEM FOR LOCAL GOVERNMENT UNITS

A Capstone Project Proposed and
Submitted to the Computer Studies Department
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WasteWise Installation Guide

1. Introduction

The WasteWise System is a waste collection monitoring and management platform designed for Ormoc City. The system is accessible through both web and mobile platforms to support System Administrators, ENRO Staff, Garbage Collectors, Barangay Officials, and Residents.

The system consists of:

- A Web Application hosted on Vercel
- A Backend Server hosted on Render
- A Mobile Application developed using React Native
- Real-time communication using WebSocket technology

The deployment setup allows users to access operational dashboards, truck tracking, route monitoring, notifications, and reporting tools using desktop computers and Android mobile devices.

2. System Requirements

2.1 Web Application Requirements

Recommended browsers:

1. Google Chrome (Latest Version)
2. Microsoft Edge
3. Minimum Requirements:
4. Stable internet connection
5. Desktop or laptop computer
6. Updated web browser

2.2 Mobile Application Requirements

Supported Platform:

1. Android Devices
2. Minimum Requirements:
3. Android Version 8.0 and above
4. At least 3 GB RAM
 - 200 MB free storage space
5. *Stable internet connection

3. WasteWise Web Application Access

The WasteWise Web Application is used by:

1. System Administrators
2. ENRO Staff
3. Barangay Officials

Web Application URL

Access the system using the following link:

<https://waste-wise-ormoc-ddm.vercel.app/>

Steps to Access the Web Application

1. Open a web browser.
2. Enter the official WasteWise URL.
3. Wait for the login page to load.
4. Enter your registered Email and Password.
5. Enter the One-Time Password (OTP) sent to your email.
6. Click Verify to access the system dashboard.

4. Backend Server Deployment Information

The WasteWise backend server is deployed using Render.

1. The backend server handles the following:
2. API requests
3. Database connections
4. Real-time WebSocket communication
5. Authentication and OTP verification
6. Data synchronization
7. Complaint and report management

Purpose of the Backend Server:

The backend server ensures smooth communication between the web application and mobile application. It enables real-time updates for truck tracking, route completion, attendance monitoring, and report notifications.

5. WasteWise Mobile Application Installation

The WasteWise Mobile Application is designed for:

- **Garbage Collectors**
- **Residents**

The application is distributed as an APK file for Android devices.

6. Downloading the APK File

The WasteWise APK file can be downloaded through:

Direct APK distribution

Google Drive link:

https://drive.google.com/drive/folders/1_revsnfxy2QukNsfrXYOmoXKTEomaCwb?usp=sharing

Steps to Download the APK

1. Open the provided Google Drive link.
2. Locate the latest WasteWise APK file.
3. Tap Download.
4. Wait for the download process to complete.

Note:

Depending on your device settings, Android may display a warning before downloading APK files from external sources.

7. Enabling Installation from Unknown Sources

Because the application is installed outside the Google Play Store, Android devices may require permission to install applications from unknown sources.

Steps to Enable Installation Permission

1. Open your Android device Settings.
2. Navigate to:
Security → Install Unknown Apps.
3. Select the browser or file manager used to download the APK.
4. Enable:
Allow from this source
5. Return to the downloaded APK file.

8. Installing the WasteWise Mobile Application

Step-by-Step Installation

1. Open the Downloads folder or File Manager.
2. Locate the WasteWise APK file.
3. Tap the APK file.

4. Tap Install.
5. Wait for the installation process to complete.
6. Tap Open to launch the application.

System Result:

The WasteWise mobile application is successfully installed on the Android device.

9. Logging In to the Mobile Application**Garbage Collector Login**

1. Open the WasteWise mobile application.
2. Enter your Email and Password.
3. Tap Login.
4. Check your email for the OTP.
5. Enter the OTP.
6. Tap Verify OTP.

Resident Access

Residents can access the tracking and reporting features directly through the mobile application.

Features Available:

1. Track garbage trucks
2. View collection logs
3. Submit complaints
4. Monitor schedules and routes

10. Updating the Mobile Application**To update the WasteWise mobile application:**

1. Download the latest APK version from the provided Google Drive link.
2. Open the updated APK file.
3. Tap Install.
4. The system will automatically replace the older version while keeping user data intact.

Note:

Users are advised to install the latest version to receive new features, bug fixes, and security improvements.

11. Troubleshooting Guide

Problem: Cannot Access the Website**Possible Causes:**

1. No internet connection
2. Incorrect URL
3. Server maintenance

Solutions:

- Check internet connectivity
- Re-enter the official URL
- Contact the system administrator

Problem: OTP Not Received**Possible Causes:**

- Incorrect email address
- Delayed email delivery
- Poor internet connection

Solutions:

- Verify the registered email address
- Check Spam or Junk folders
- Request a new OTP

Problem: APK Installation Blocked**Possible Causes:**

- Unknown Sources permission disabled

Solutions:

- Enable installation from unknown sources in Android settings

Problem: Application Not Opening**Possible Causes:**

- Unsupported Android version
- Corrupted APK file

Solutions:

- Re-download the APK file
- Update Android software if possible
- Restart the device and try again

12. Security Reminders

1. To ensure account security:
2. Do not share your password or OTP.
3. Always log out after using the system.
4. Install only official APK files provided by authorized personnel.
5. Regularly update the application.

WasteWise User Manual

Monitor. Manage. Maintain.

From Collection to Operation.

Together, we track, act, and help keep Ormoc City clean.

=====ACCESSING THE WASTEWISE SYSTEM (WEB APPLICATION)=====

The WasteWise Web Application is used by **System Administrators** and **ENRO Staff** to manage waste collection operations.

Available Actions on the Login Page

- **Login** – Access the system using your credentials
- **Request Account Access** – For eligible users only
- **Forgot Password** – Reset a forgotten password

1. Request Account Access (*ENRO Staff and Barangay Officials Only*)

Button Used: Request Account Access

Step-by-Step Instructions:

1. Open the WasteWise login page.
2. Click the **Request Account Access** button.
3. Enter the following details:
 - Full Name
 - Gender
 - Mobile Number
 - Email Address
 - Role(s)
4. Click **Submit Request**.

System Action:

- The request is sent to the **System Administrator** for review.

If Approved:

1. Open your email.
2. Read the approval notification.
3. Return to the login page.

4. Enter your **Email** and **Password**.
5. Enter the **OTP** sent to your email.
6. Click **Verify** to access the system.

===== FORGOT PASSWORD (WEB APPLICATION) =====

2. Button Used: **Forgot Password**

Step-by-Step Instructions:

1. On the login page, click **Forgot Password**.
2. Enter your registered email address.
3. Click **Send Code**.
4. Check your email for the OTP.
5. Enter the OTP in the system.
6. Create a new password.
7. Click **Save Password**.
8. Click **Login** to access your account.

3. Mobile Application Access Overview

The WasteWise Mobile Application supports two user types:

- **Garbage Collector**
- **Resident**

Each user has different buttons, pages, and actions.

===== GARBAGE COLLECTOR USER MANUAL (MOBILE APP) =====

4.1 Login (Garbage Collector)

Buttons Used: **Login**, **Verify OTP**

Step-by-Step Instructions:

1. Open the WasteWise mobile app.
2. Enter your **Email** and **Password**.
3. Tap **Login**.
4. Check your email for the OTP.
5. Enter the OTP.
6. Tap **Verify OTP** to proceed.

4.2 Dashboard

The dashboard displays your daily work overview.

4.3 Attendance Page (Before Starting Collection)

Buttons Used: [Clock In](#), [Clock Out](#)

Step-by-Step Instructions:

1. Open the **Attendance Page**.
2. Tap **Clock In** to start your shift.
3. Confirm the action when prompted.

System Result:

- Status changes to **On Duty**.
- The button changes to **Clock Out**.
- ENRO Staff and Residents can now see that collection has started.

4.4 Schedule Page

Button Used: [View Schedule](#)

Step-by-Step Instructions:

1. Tap the **Schedule Page**.
2. View your assigned route for the day.
3. Scroll to see upcoming schedules.

4.5 Route Page

Buttons Used: [Map View](#), [Check](#)

Step-by-Step Instructions:

1. Open the **Route Page**.
2. View the highlighted route line on the map.
3. Follow the route from **Start Point** to **End Point**.
4. When a collection point is completed, tap **Check**.

System Result:

- Status changes from **Pending** to **Completed**.

- Updates are reflected in real time across all systems.

4.6 Report Page (Garbage Collector)

Buttons Used: [Report](#), [Submit Report](#)

Step-by-Step Instructions:

1. Open the **Report Page**.
2. Tap **Report**.
3. Select a **Report Type**:
 - Vehicle Issue
 - Equipment Problem
 - Route Issue
 - Safety Incident
4. Select the **Specific Issue**.
5. Enter notes or remarks.
6. Tap **Submit Report**.

System Result:

- ENRO Staff are automatically notified.

4.7 Settings Page

Buttons Used: [Edit Profile](#), [Login History](#)

Available Actions:

- Update personal information
- View login activity (device, time, status)

===== RESIDENT USER MANUAL (MOBILE APP) =====

5.1 Track Page

Buttons Used: [Search](#), [Filter](#)

Step-by-Step Instructions:

1. Open the WasteWise mobile app.
2. The **Track Page** opens by default.
3. Tap **Search** to enter:

- Truck ID
- Route Name
- Barangay
- 4. Tap **Filter** to select:
 - Truck Status
 - Garbage Type
 - Schedule Day
- 5. Apply filters to update the map view.

5.2 Logs Page

Buttons Used: View Logs, Filter

Step-by-Step Instructions:

1. Tap the **Logs Page**.
2. View collection status (**Completed** / **Incomplete**).
3. Use **Filter** to refine results.

5.3 Report Page (Resident Complaint)

Buttons Used: Submit Complaint

Step-by-Step Instructions:

1. Open the **Report Page**.
2. Tap **Submit Complaint**.
3. Select complaint type:
 - Uncollected Garbage
 - Overflowing Garbage
 - Illegal Dumping
 - Missed Route
 - Others
4. Select garbage type.
5. Enter notes.
6. Tap **Submit**.

System Result:

- Complaint is sent directly to ENRO Staff.

6. Forgot Password (Mobile App – Garbage Collector)

Button Used: **Forgot Password**

Step-by-Step Instructions:

1. On the login screen, tap **Forgot Password**.
2. Enter your registered email address.
3. Tap **Send Code**.
4. Enter the OTP received via email.
5. Create a new password.
6. Tap **Save**.

===== **SYSTEM ADMINISTRATOR USER MANUAL** =====

7.1 User Management

Buttons Used: **Add New User**, **Save**

Step-by-Step Instructions:

1. Open **User Management**.
2. Click **Add New User**.
3. Enter user details.
4. Assign role(s).
5. Click **Save**.

System Result:

- Credentials are emailed to the user.

7.2 Role Action Management

Buttons Used: **Add Role Action**, **Save**

Step-by-Step Instructions:

1. Open **Role Action Management**.
2. Click **Add Role Action**.
3. Select role.
4. Set permissions and capabilities.
5. Click **Save**.

===== **ENRO STAFF USER MANUAL** =====

8.1 Barangay Management

Buttons Used: [Barangay Management](#), [Add Barangay](#), [Save](#)

Step-by-Step Instructions:

1. Navigate to **Barangay Management** from the sidebar.
2. View the list of all barangays in Ormoc City.
3. Click **Add Barangay** if a new barangay needs to be added.
4. Enter the barangay details.
5. Click **Save**.

Purpose:

- Barangays serve as references for **route creation** and **schedule assignment**.

8.2 Truck Management

Buttons Used: [Truck Management](#), [Add New Truck](#), [Save](#)

Step-by-Step Instructions:

1. Open **Truck Management**.
2. Click **Add New Truck**.
3. Enter the following information:
 - Truck ID
 - Truck Status (Active, On Route, Inactive, Under Maintenance)
 - Assign Garbage Collector
4. Click **Save**.

System Result:

- The truck becomes available for route and schedule assignment.

8.3 Route Management

Buttons Used: [Route Management](#), [Add New Route](#), [Add](#), [Save Route](#), [Clear Route](#)

Step-by-Step Instructions:

1. Navigate to **Route Management**.
2. Click **Add New Route**.

3. Enter the **Route Name**.
4. Select one or multiple **Barangays**.
5. Choose a **Polyline Color** for the route.
6. On the map, click to define the **Start Point**.
7. Continue clicking to draw the **Complete Route Path**.
8. Click the last point to set the **End Point**.
9. Click **Save Route**.

Purpose:

- Defines the official route followed by garbage trucks.
- Routes are displayed to **Garbage Collectors** and **Residents**.

8.4 Schedule Management

Buttons Used: [Schedule Management](#), [Add New Schedule](#), [Save](#)

Step-by-Step Instructions:

1. Open **Schedule Management**.
2. Click **Add New Schedule**.
3. Select a **Route** (automatically loaded from Route Management).
4. Assign a **Garbage Collector**.
5. Verify the **Truck Status** (auto-linked).
6. Select the **Garbage Type**.
7. Choose the **Collection Days**.
8. Click **Save**.

System Result:

- The schedule becomes active and visible to collectors and residents.

8.5 Garbage Report Management

Buttons Used: [Garbage Report Management](#), [View](#), [Update Status](#)

Step-by-Step Instructions:

1. Navigate to **Garbage Report Management**.
2. View complaints submitted by residents.
3. Select a report to view details:
 - Barangay
 - Report content

- Date submitted
- 4. Update the complaint status (e.g., Pending, In Progress, Resolved).
- 5. Save changes.

System Result:

- Residents can see updates on their submitted complaints.

8.6 Garbage Site Management

Buttons Used: Garbage Site Management, Add New Garbage Site, Save

Step-by-Step Instructions:

1. Open **Garbage Site Management**.
2. Click **Add New Garbage Site**.
3. Select the **Barangay**.
4. Enter the **Site Name**.
5. Use the map to mark the **Drop-off Point**.
6. Click **Save**.

System Result:

- Garbage drop sites are visible to residents in the mobile app.

8.7 Collector Report Management

Buttons Used: Collector Report Management, View, Update Status

Step-by-Step Instructions:

1. Navigate to **Collector Report Management**.
2. View reports submitted by garbage collectors.
3. Open a report to see the issue details.
4. Update the issue status (Resolved / Unresolved).
5. Save the update.

8.8 Collector Attendance Management

Buttons Used: Collector Attendance Management, View

Step-by-Step Instructions:

1. Open **Collector Attendance Management**.

2. Monitor collector attendance status:
 - Clocked In (Started Route)
 - Clocked Out (Completed Route)
3. Use this information for operational monitoring.

8.9 Truck Map

Buttons Used: [Truck Map](#), [Search](#), [Filter](#)

Step-by-Step Instructions:

1. Open **Truck Map**.
2. View all trucks and routes displayed on the map.
3. Use **Search** to find a specific truck, route, or barangay.
4. Use **Filter** to refine results by:
 - Truck Status
 - Route
 - Barangay

Purpose:

- Provides real-time visibility of garbage collection operations.

8.10 Profile and Login History

Buttons Used: [Profile](#), [Edit Profile](#), [Change Password](#), [Login History](#)

Step-by-Step Instructions:

1. Open **Profile**.
2. Click **Edit Profile** to update personal information.
3. Click **Change Password** to set a new password.
4. Open **Login History** to view: (Login timestamps, Device information, Login status)